

```
Instance")
shap.plots.waterfall(shap_values[0], show=False)
plt.title("Waterfall Plot: SHAP Value Breakdown for First Prediction", fontsize=14)
plt.tight_layout()
plt.show()

# 2. Beeswarm plot (global)
plt.figure(figsize=(10, 6))
print("[2] Beeswarm Plot: Global SHAP Feature Importance")
shap.plots.beeswarm(shap_values, show=False)
plt.title("Beeswarm Plot: Global Feature Impact",
          fontsize=14)
plt.xlabel("SHAP Value (Effect on Prediction Output)",
          fontsize=12)
plt.ylabel("Features", fontsize=12)
plt.tight_layout()
plt.show()

# 3. Force plot (local)
print("[3] Force Plot: Push and Pull Effect for First Instance")
shap.plots.force(shap_values[0], matplotlib=True)
```

Figure 3 shows the program output.

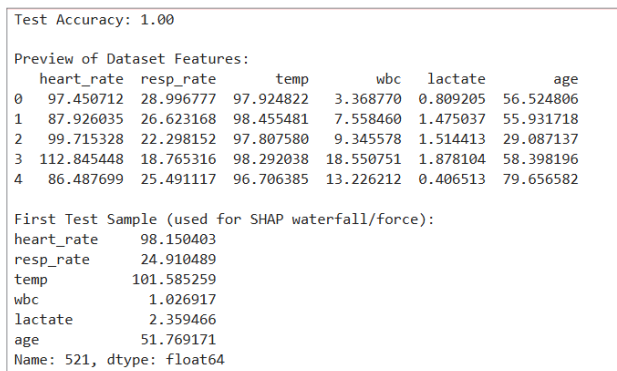


Figure 3: Program output

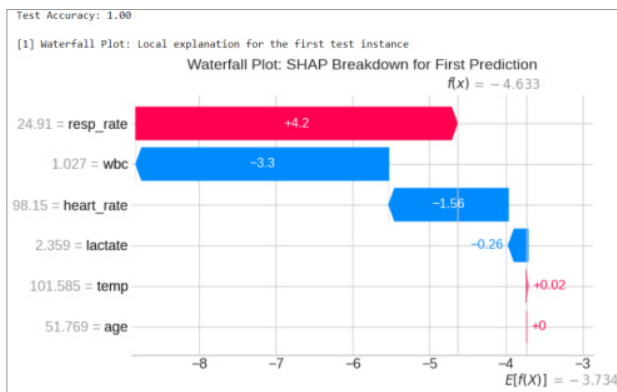


Figure 4: SHAP waterfall plot

Explainable AI with waterfall, beeswarm, and force plots

An XGBoost model is trained to predict whether a patient is at high risk of sepsis based on clinical vitals (heart rate, respiration, WBC count, etc). Since decisions in healthcare have high stakes, explainability is not optional — it is critical. SHAP (SHapley Additive exPlanations) provides transparent and mathematically grounded explanations that help:

- Understand individual predictions (local interpretability)
- Understand global model behaviour (global interpretability)

Waterfall plot: The SHAP waterfall plot gives a local explanation for the prediction for one individual patient. It visually breaks down how each feature pushed the prediction up or down from the model's baseline.

Let's dive into this plot in the context of explainable AI in healthcare (sepsis prediction).

- The model starts with a baseline prediction ($E[f(x)]$) of -3.734 , which is the average prediction across all patients.
- Then, each feature's SHAP value adds or subtracts from this base to reach the final prediction $f(x) = -4.633$.
- Arrows show positive (red) or negative (blue) influence on the prediction.

	Feature	Value
	Respiration rate	24.91
	WBC count	1.03
	Heart rate	98.15
	Lactate level	2.36
	Temperature	101.59
	Age	51.77

Waterfall SHAP explanation

	Feature	Value	SHAP value	Impact on prediction
	resp_rate	24.91	+4.2	Strongly increased sepsis risk
	wbc	1.03	-3.3	Strongly decreased risk
	heart_rate	98.15	-1.56	Slightly decreased risk
	lactate	2.36	-0.26	Slight decrease
	temp	101.59	+0.02	Neutral
	age	51.77	0.00	No impact